Experiment No: 3_2

Title: Establish the Connection Between Two Devices Using One Switch Through Copper Straight-Through Cable

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Aim

To establish a connection between 2 devices using switch though copper straight through cable

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Requirements

Hardware Requirements

- PC-A
- PC-B
- One Switch
- Copper Straight-Through Cables

Software Requirements

- Cisco Packet Tracer / Network Simulator

Physical Connection

Device	Connected To	Switch Port	Cable Type
PC-A	Switch	Fa0/1	Copper Straight-Through
PC-B	Switch	Fa0/2	Copper Straight-Through

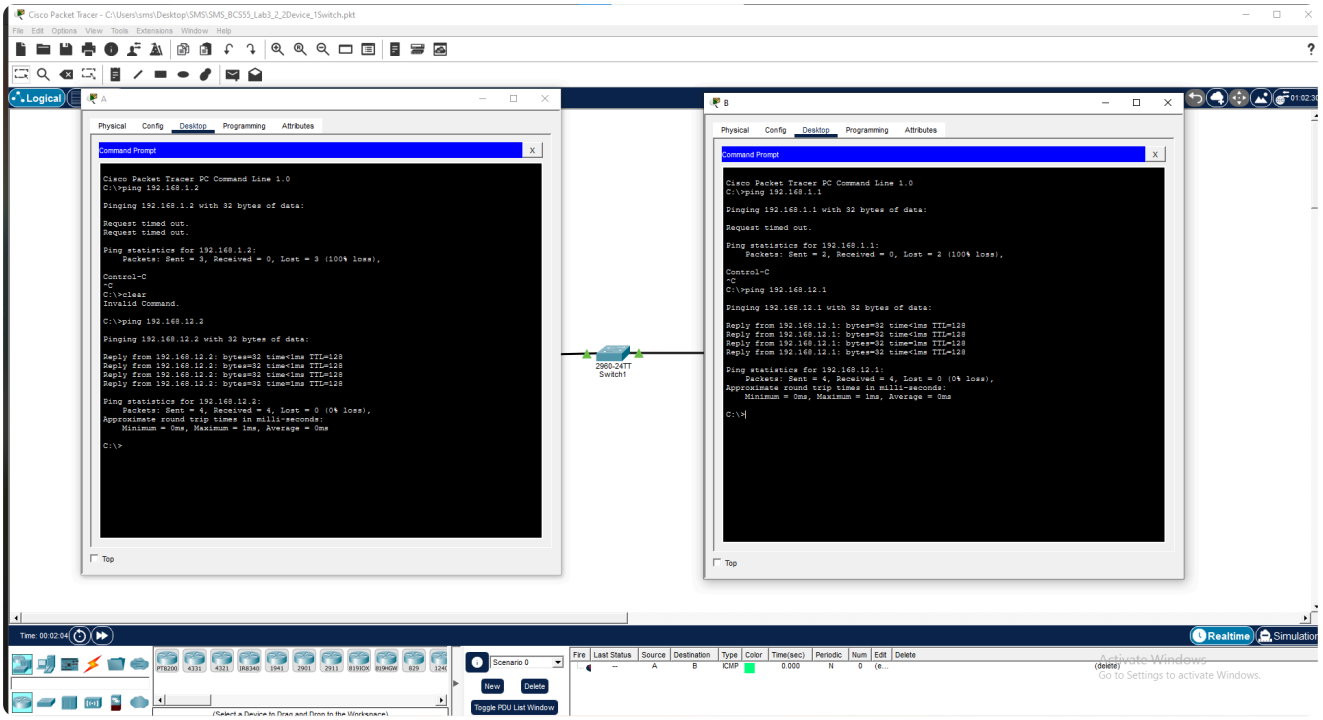
Procedure

1. Open Cisco Packet Tracer.
2. Place PCs (PC-A, PC-B) and one switch in the workspace.
3. Connect all PCs to the switch using Copper Straight-Through cables.
4. Assign IP addresses to all PCs manually.
5. Verify IP configuration using `ipconfig` / `ifconfig`.
6. Test connectivity between all devices using `ping`.
7. Switch to **Simulation Mode** and observe packet transmission.
8. Verify subnet mask configuration on all devices.

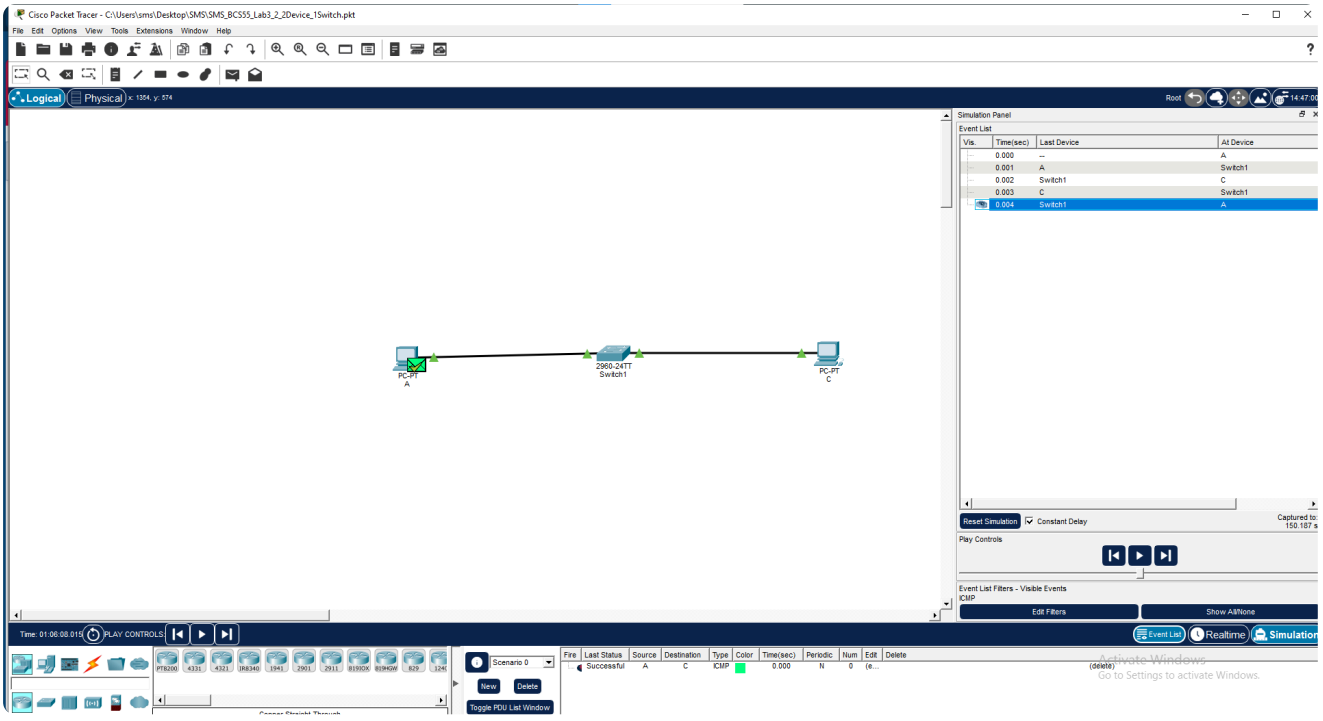
Expected Output (Screenshots)

1. IP Address of Each Device and IP Class

Device	IP Address	Subnet Mask	IP Class
PC-A	192.168.12.1	255.255.255.0	C
PC-B	192.168.12.1	255.255.255.0	C



Status of Packets in Simulation Panel



Observations

- All devices were successfully connected through the switch.
- IP addresses were assigned correctly and belonged to the same network class.
- Ping responses confirmed end-to-end connectivity.
- Packet flow was successful in Simulation Mode.

Result

Thus, the connection between two devices using one switch through Copper Straight-Through cables was successfully established and verified.

Conclusion

This experiment demonstrates basic LAN configuration using a switch, IP addressing, and verification of network communication between multiple devices.